



Modelling Financial Markets

Seminar 6: Multivariate Volatility and Dependence — Hands-on



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ BAC: Bank of America Corporation (NYSE ticker)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BUX: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ CCC: Constant Conditional Correlation (model)
- ▣ CCC-NL: Constant Conditional Correlation with NonLinear shrinkage
- ▣ CML: Canonical Maximum Likelihood
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DCC: Dynamic Conditional Correlation
- ▣ DCC-NL: Dynamic Conditional Correlation with NonLinear shrinkage
- ▣ DCC-S: Dynamic Conditional Correlation with the Sample correlation target
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GAS: Generalised Autoregressive Score (model)
- ▣ GMV: Global Minimum-Variance portfolio
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ JPM: JPMorgan Chase & Co. (NYSE ticker)
- ▣ SD: Standard Deviation
- ▣ STOXX50E: Euro Stoxx 50 index (symbol of the data file)
- ▣ US: United States
- ▣ WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Outline

Introduction

Part A: Derivations

Part B: Estimation and Inference

Part C: Open Question

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Compute tail dependence and copula parameters step by step
- ▣ Estimate a DCC model, test it against constant correlation and interpret its persistence
- ▣ Fit and test copulas on pseudo-observations with a parametric bootstrap
- ▣ Separate contagion from the volatility bias (Forbes–Rigobon)

- ▣ Background: Lecture 6; textbooks [Franke, Härdle & Hafner \(2019\)](#) and [Nelsen \(2006\)](#)

Structure

- ▣ **Part A:** six derivations on paper (A1–A6)
- ▣ **Part B:** six computational exercises with inference (B1–B6)
- ▣ **Part C:** one open question, a seed for the team project (C1), a critique of an AI answer (C2) and a project seed from a landmark paper (C3)

How to Work in This Seminar

- ▣ **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- ▣ Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions: ask what would change the conclusion
- ▣ Report every estimate with its uncertainty: standard error, bootstrap interval or p -value
- ▣ Ask each time whether the standard method can fail here: time zones, boundary parameters, volatility changes, many observations

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (deriving the tail dependence of the t copula), A2 (Kendall's tau, copula parameters and their standard errors), A3 (Forbes–Rigobon adjustment with an estimated δ)
- Part B: B1 (DCC on SPY and TLT, break at an unknown date, two-step standard errors), B2 (Gaussian versus t copula, sandwich standard errors)

Proposed: you solve them

- Part A: A4 (model: A3), A5 (model: B1), A6 (model: A2)
- Part B: B3 (model: B2, A2), B4 (model: A3), B5 (model: B1), B6 (model: B2)
- Part C: C1, open question and project seed; C2, critique of an AI answer (model: A1, A2); C3, large dynamic covariance matrices (model: B1, C1)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter6_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `arch`, `matplotlib`
- Series: SPY, TLT, S&P 500, Euro Stoxx 50, BET, Bitcoin; daily closes up to 18 September 2026
- Joint analyses: prices are first joined on common trading days, then returns are computed
- Weekly returns: last common close of each week (Friday); 2-day returns: $\ln P_t - \ln P_{t-2}$

Formulas Used Today

- DCC: $Q_t = (1 - a - b)\bar{Q} + a\varepsilon_{t-1}\varepsilon_{t-1}^\top + bQ_{t-1}$, $R_t = \text{diag}(Q_t)^{-1/2}Q_t\text{diag}(Q_t)^{-1/2}$ Engle (2002)
- Forbes–Rigobon: $\delta = \sigma_{x,c}^2/\sigma_{x,calm}^2 - 1$, $\rho^* = \rho_c/\sqrt{1 + \delta(1 - \rho_c^2)}$ Forbes & Rigobon (2002)
- Pseudo-observations: $\hat{u}_{it} = \text{rank}(x_{it})/(T + 1)$ Genest, Ghoudi & Rivest (1995)
- Tail dependence: $\lambda_L = \lim_{q \downarrow 0} C(q, q)/q$; t copula: $\lambda = 2t_{\nu+1}(-\sqrt{(\nu+1)(1-\rho)/(1+\rho)})$ Demarta & McNeil (2005)
- Kendall's tau: Gaussian/t $\tau = \frac{2}{\pi} \arcsin \rho$; Clayton $\theta/(\theta + 2)$; Gumbel $1 - 1/\theta$
- Goodness of fit: Cramér–von Mises distance of the Rosenblatt pairs $(u, h(v | u))$ from independence, parametric bootstrap Genest, Rémillard & Beaudoin (2009)

A1: Tail Dependence of the t Copula: Task (1/2) [Solved]

Context and data

- No data are needed for (a)–(c); (d) and (e) use $\rho = 0.5$ and $\nu \in \{3, 4, 10, 30\}$
- Definition: $\lambda_L = \lim_{q \downarrow 0} C(q, q)/q$; the copula is exchangeable, $C(u, v) = C(v, u)$
- Fact: for a bivariate Student t with correlation ρ and ν degrees of freedom, given $X_1 = x$ the ratio $(X_2 - \rho x) / \sqrt{(\nu + x^2)(1 - \rho^2)/(\nu + 1)}$ follows $t_{\nu+1}$

What to do

- (a) Use L'Hôpital's rule to show that $\lambda_L = 2 \lim_{q \downarrow 0} P(V \leq q \mid U = q)$
- (b) Substitute the conditional distribution of the t copula and let $x = t_{\nu}^{-1}(q) \rightarrow -\infty$ to derive $\lambda = 2 t_{\nu+1}(-\sqrt{(\nu + 1)(1 - \rho)/(1 + \rho)})$
- (c) Repeat the argument for the Gaussian copula and show that $\lambda = 0$ whenever $|\rho| < 1$

A1: Tail Dependence of the t Copula: Task (2/2) [Solved]

What to do

- (d) Evaluate λ of the t copula with $\rho = 0.5$ for each ν
- (e) Compute the finite-level probability $P(V \leq q \mid U \leq q) = C(q, q)/q$ at $q = 5\%$ and $q = 0.1\%$ for the Gaussian copula and for the t copula with $\nu = 4$, from the bivariate distribution functions

What to report and interpret

- The derivation of (a)–(c) as a chain of equalities, each step justified in one sentence
- A table of λ by ν and the four finite-level probabilities of (e)
- Interpretation: why can a Gaussian risk model calibrated at the 5% level miss joint crashes at the 0.1% level?

A1: Tail Dependence of the t Copula: Solution (1/2) [Solved]

Solution

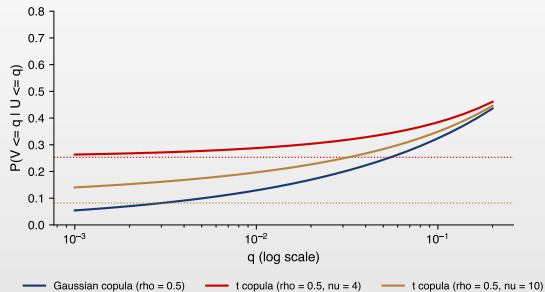
- (a) Both numerator and denominator of $C(q, q)/q$ tend to 0, so by L'Hôpital $\lambda_L = \lim_{q \downarrow 0} \frac{d}{dq} C(q, q)$
- By the chain rule $\frac{d}{dq} C(q, q) = \partial_1 C(q, q) + \partial_2 C(q, q)$, and $\partial_1 C(u, v) = P(V \leq v \mid U = u)$; exchangeability makes the two terms equal, so $\lambda_L = 2 \lim_{q \downarrow 0} P(V \leq q \mid U = q)$
- (b) With $x = t_\nu^{-1}(q)$: $P(V \leq q \mid U = q) = P(X_2 \leq x \mid X_1 = x) = t_{\nu+1} \left(\frac{(1-\rho)x}{\sqrt{(\nu+x^2)(1-\rho^2)/(\nu+1)}} \right)$
- As $q \downarrow 0$, $x \rightarrow -\infty$ and $x/\sqrt{\nu+x^2} \rightarrow -1$; since $(1-\rho)/\sqrt{1-\rho^2} = \sqrt{(1-\rho)/(1+\rho)}$, the argument tends to $-\sqrt{(\nu+1)(1-\rho)/(1+\rho)}$, which gives the formula
- (c) Gaussian: $P(V \leq q \mid U = q) = \Phi(x\sqrt{(1-\rho)/(1+\rho)})$ with $x = \Phi^{-1}(q) \rightarrow -\infty$, so the probability tends to 0 and $\lambda = 0$
- Interpretation: in the t case the conditional scale grows with $|x|$, so the standardised argument stays bounded; in the Gaussian case the scale is fixed and the argument diverges

A1: Tail Dependence of the t Copula: Solution (2/2) [Solved]

Solution

- (d) Formula: $\lambda = 2 t_{\nu+1}(-a_\nu)$ with $a_\nu = \sqrt{(\nu+1)(1-\rho)/(1+\rho)} = \sqrt{(\nu+1) \cdot 0.5/1.5}$
- Substituted values: $a_3 = 1.155$, $a_4 = 1.291$, $a_{10} = 1.915$, $a_{30} = 3.215$
- Intermediate result for $\nu = 4$: $t_5(-1.291) = 0.1266$, hence $\lambda = 2 \times 0.1266 = 0.253$
- All values: $\lambda = 0.313$ ($\nu = 3$), 0.253 ($\nu = 4$), 0.082 ($\nu = 10$), 0.003 ($\nu = 30$): tail dependence vanishes as ν grows
- (e) Gaussian: 0.244 at 5% and 0.054 at 0.1%; t with $\nu = 4$: 0.339 and 0.263
- Interpretation: at 5% the two models look alike, but at 0.1% the Gaussian probability has fallen to a fraction of the t one; a model calibrated on moderate losses cannot reveal this

A1: The Picture [Solved]



- Exact $P(V \leq q | U \leq q)$ from the bivariate Normal and t distribution functions; dotted lines: the limits λ
- Read the chart from right to left: as q falls, the Gaussian curve heads to 0 and the t curves flatten at their limits λ

A2: Kendall's Tau, Copula Parameters and Their Precision: Task (1/2) [Solved]

Context and data

- Data: weekly log returns of the S&P 500 (GSPC.INDX) and the Euro Stoxx 50 (STOXX50E.INDX), Friday closes on common trading days, 2000–2026, $n = 1393$
- Pseudo-observations $\hat{u}_i = \text{rank} / (n + 1)$; weekly pairs treated as i.i.d.
- Sample values: Kendall's $\hat{\tau} = 0.564$, Spearman's $\hat{\rho}_S = 0.746$, Pearson $\hat{\rho} = 0.781$

What to do

- (a) Invert $\hat{\tau}$ for the Gaussian ($\rho = \sin(\pi\tau/2)$), Clayton ($\theta = 2\tau/(1 - \tau)$), Gumbel ($\theta = 1/(1 - \tau)$) and Frank (solve $1 - \frac{4}{\theta}[1 - D_1(\theta)] = \hat{\tau}$ numerically) copulas
- (b) Compute the tail dependence implied by each fitted copula
- (c) Compute Spearman's rho implied by the Gaussian copula, $\rho_S = \frac{6}{\pi} \arcsin(\rho/2)$, and compare it with the sample value

A2: Kendall's Tau, Copula Parameters and Their Precision: Task (2/2) [Solved]

What to do

- (d) From the Hoeffding projection $W_i = 2\hat{C}_n(\hat{U}_i, \hat{V}_i) - \hat{U}_i - \hat{V}_i$ (sample sd 0.1239), show that $\text{se}(\hat{\tau}) = 4 \text{sd}(W)/\sqrt{n}$ and compute it; then the delta-method se of $\hat{\theta}_{\text{Clayton}}$ and $\hat{\rho}_{\text{Gauss}}$
- (e) Decide whether the gap in (c) is sampling error with an i.i.d. bootstrap of the weekly pairs (1999 draws, 95% percentile interval)

What to report and interpret

- A table with the four parameters, their tail dependence and the standard errors of (d)
- The gap of (c) with its bootstrap interval and a decision at the 5% level
- Interpretation: can $\hat{\tau}$ alone choose the copula family? Can the pair $(\hat{\tau}, \hat{\rho}_S)$ reject one?

A2: Kendall's Tau, Copula Parameters and Their Precision: Solution (1/2) [Solved]

Solution

- (a) Gaussian: $\rho = \sin(\pi \times 0.564/2) = 0.775$
- Clayton: $\theta = 2 \times 0.564/0.436 = 2.589$; Gumbel: $\theta = 1/0.436 = 2.295$; Frank: the numerical root is $\theta = 7.044$
- (b) Clayton: $\lambda_L = 2^{-1/2.589} = 0.765$; Gumbel: $\lambda_U = 2 - 2^{1/2.295} = 0.647$; Gaussian and Frank: no tail dependence
- The same $\hat{\tau}$ gives tail dependence between 0 and 0.77, depending only on the family chosen
- (c) $\frac{6}{\pi} \arcsin(0.775/2) = 0.760$ against the sample 0.746: a gap of 0.0134, small in absolute terms; (e) checks whether it is noise

A2: Kendall's Tau, Copula Parameters and Their Precision: Solution (2/2) [Solved]

Solution

- (d) $\hat{\tau}$ is a U-statistic with kernel h ; $E[h \mid Z_i] = 4C(U_i, V_i) - 2U_i - 2V_i + 1 = 2W_i + 1$, so Hoeffding gives $\text{Var}(\hat{\tau}) \approx 4\text{Var}(2W)/n = 16\text{Var}(W)/n$
- Substitution: $\text{se}(\hat{\tau}) = 4 \times 0.1239 / \sqrt{1393} = 0.0133$
- Clayton: $d\theta/d\tau = 2/(1-\tau)^2 = 10.53$, so $\text{se} = 10.53 \times 0.0133 = 0.140$; Gaussian: $d\rho/d\tau = \frac{\pi}{2} \cos(\pi\tau/2) = 0.993$, so $\text{se} = 0.0132$ (Genest & Rivest (1993))
- (e) Bootstrap: gap 0.0134, se 0.0026, 95% interval [0.0088, 0.0187]; zero is outside, so the gap is not sampling error at 5%
- Why so precise: $\hat{\tau}$ and $\hat{\rho}_S$ move together across samples, so their difference has a much smaller se than each of them
- Interpretation: $\hat{\tau}$ alone fits every family, but the pair $(\hat{\tau}, \hat{\rho}_S)$ already rejects the Gaussian copula for these data

A3: The Forbes–Rigobon Adjustment with an Estimated δ : Task (1/2) [Solved]

Context and data

- Model: $y_t = \alpha + \beta x_t + \varepsilon_t$ with β and σ_ε^2 constant; only the variance of the source market x changes in the crisis, $\sigma_{x,c}^2 = (1 + \delta)\sigma_x^2$
- Stylised numbers for (a)–(c): calm correlation 0.40, crisis correlation 0.60, source variance 4 times higher in the crisis
- Data for (d)–(e): 2-day returns of the S&P 500, Euro Stoxx 50 and BET, calm 14 Sep 2007 – 12 Sep 2008, crisis 15 Sep 2008 – 31 Mar 2009; $\hat{\delta} = 6.54$, effective $n_c = 65$ and $n_0 = 121$ (half the overlapping 2-day returns, an approximation)

What to do

- (a) Compute δ and the adjusted crisis correlation $\rho^* = \rho_c / \sqrt{1 + \delta(1 - \rho_c^2)}$
- (b) Compute the crisis correlation we would measure if the true correlation stayed at 0.40, using $\rho_c = \rho \sqrt{(1 + \delta)/(1 + \delta\rho^2)}$
- (c) Decide between contagion and interdependence

A3: The Forbes–Rigobon Adjustment with an Estimated δ : Task (2/2) [Solved]

What to do

- (d) Under i.i.d. Normality $(1 + \hat{\delta})/(1 + \delta) \sim F(n_c - 1, n_0 - 1)$; derive $\text{Var}(\hat{\delta})$ to first order
- (e) By the delta method in $(\hat{\rho}_c, \hat{\delta})$, including their covariance (both use the crisis window), compute $\text{se}(\rho^*)$ with δ fixed and with δ random, for Euro Stoxx 50 and BET in 2008

What to report and interpret

- The numbers of (a)–(b), each with the formula and the substituted values
- $\text{se}(\hat{\delta})$, the two standard errors of ρ^* and their percentage difference
- Interpretation: does treating δ as random change the 2008 verdict of no contagion?

A3: The Forbes–Rigobon Adjustment with an Estimated δ : Solution (1/2) [Solved]

Solution

- (a) Variance ratio 4, so $\delta = 4 - 1 = 3$
- $\rho^* = 0.60 / \sqrt{1 + 3(1 - 0.36)} = 0.60 / \sqrt{2.92} = 0.60 / 1.709 = 0.351$
- (b) $\rho_c = 0.40 \sqrt{(1 + 3) / (1 + 3 \times 0.16)} = 0.40 \sqrt{4 / 1.48} = 0.40 \times 1.644 = 0.658$
- (c) The adjusted correlation 0.351 is below the calm 0.40, and volatility alone would have lifted 0.40 to 0.658, more than the observed 0.60
- Conclusion: the whole rise is explained by higher volatility of the source market, i.e. interdependence, not contagion

A3: The Forbes–Rigobon Adjustment with an Estimated δ : Solution (2/2) [Solved]

Solution

- ▣ (d) $\text{Var}[\ln \chi_k^2/k] \approx 2/k$, so $\text{Var} \ln(1 + \hat{\delta}) \approx \frac{2}{n_c - 1} + \frac{2}{n_0 - 1}$ and $\text{Var}(\hat{\delta}) \approx 2(1 + \delta)^2 \left[\frac{1}{n_c - 1} + \frac{1}{n_0 - 1} \right]$
- ▣ Substitution: $\text{se}(\hat{\delta}) = 7.54 \sqrt{2(1/64 + 1/120)} = 1.65$
- ▣ (e) With $k = 1 + \delta(1 - \rho_c^2)$: $g_r = \partial \rho^* / \partial \rho_c = (1 + \delta)k^{-3/2}$, $g_\delta = \partial \rho^* / \partial \delta = -\rho_c(1 - \rho_c^2)k^{-3/2}/2$; $\text{Var}(\hat{\rho}_c) \approx (1 - \rho_c^2)^2/n_c$
- ▣ Bivariate Normal: $\text{Cov}(\hat{\rho}_c, \ln \hat{\sigma}_{x,c}^2) \approx \rho_c(1 - \rho_c^2)/n_c$, so $\text{Cov}(\hat{\rho}_c, \hat{\delta}) \approx (1 + \delta)\rho_c(1 - \rho_c^2)/n_c$;
 $\text{Var}(\rho^*) \approx g_r^2 \text{Var} \hat{\rho}_c + g_\delta^2 \text{Var} \hat{\delta} + 2g_r g_\delta \text{Cov}(\hat{\rho}_c, \hat{\delta})$
- ▣ Euro Stoxx 50 ($\rho_c = 0.804$): $g_r = 1.253$, $g_\delta = -0.0236$; se 0.055 with δ fixed, 0.051 with δ random (−7%)
- ▣ BET ($\rho_c = 0.485$): se $0.049 \rightarrow 0.047$ (−3%); testing $\rho^* - \hat{\rho}_{calm}$ also uses $\text{Cov}(\hat{\delta}, \hat{\rho}_{calm}) \approx -(1 + \delta)\rho_{calm}(1 - \rho_{calm}^2)/n_0$:
 $z = -0.19$ (BET), $z = -2.44$ (Euro Stoxx 50)
- ▣ Interpretation: both adjusted correlations stay below the calm ones, so the verdict of no contagion holds; a random δ need not widen the interval: a volatile crisis window raises $\hat{\delta}$ and $\hat{\rho}_c$ together, and their errors partly cancel in ρ^*

A4: Deriving the Forbes–Rigobon Bias: Task (1/2) [Proposed]

Solution discussed in class

Context and data

- Model $y_t = \alpha + \beta x_t + \varepsilon_t$ with $\text{Cov}(x_t, \varepsilon_t) = 0$; β and σ_ε^2 are the same in the calm period and in the crisis
- In the crisis only the variance of x changes: $\sigma_{x,c}^2 = (1 + \delta)\sigma_x^2$, $\delta > 0$

What to do

- (a) Write ρ^2 as a function of $k = \beta^2 \sigma_x^2 / \sigma_\varepsilon^2$
- (b) Derive the measured crisis correlation $\rho_c = \rho \sqrt{(1 + \delta) / (1 + \delta \rho^2)}$ and show that $|\rho_c| > |\rho|$ for $0 < |\rho| < 1$, i.e. $\rho_c > \rho$ when $\rho > 0$
- (c) Invert the relation to obtain $\rho^* = \rho_c / \sqrt{1 + \delta(1 - \rho_c^2)}$
- (d) State which assumption fails if a global shock raises both σ_x and σ_ε , and in which direction the adjustment is then biased

A4: Deriving the Forbes–Rigobon Bias: Task (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- Each step of the derivation with one sentence of justification; the answer to (d) in economic terms
- Model: A3

A5: DCC Algebra: Task (1/2) [Proposed]

Solution discussed in class

Context and data

- DCC recursion $Q_t = (1 - a - b)\bar{Q} + a\varepsilon_{t-1}\varepsilon_{t-1}^\top + bQ_{t-1}$ with $a, b \geq 0$, $a + b < 1$, $\bar{Q}$ positive definite, $Q_0 = \bar{Q}$
- B1 estimates for SPY and TLT: $a = 0.0522$, $b = 0.9335$

What to do

- (a) Show by induction that Q_t is positive definite for every t
- (b) Assuming $\mathbb{E}[\varepsilon_t\varepsilon_t^\top] = \bar{Q}$ and stationarity, show that $\mathbb{E}[Q_t] = \bar{Q}$; explain why this assumption does not hold exactly for the standard DCC
- (c) Compute the approximate half-life $\ln 0.5 / \ln(a + b)$ of a shock to Q_t with the B1 estimates, say why it is only approximate for the standard DCC, and describe the case $a + b = 1$
- (d) For the corrected DCC of [Aielli \(2013\)](#), $Q_t = (1 - a - b)\bar{Q}^* + aQ_{t-1}^{*1/2}\varepsilon_{t-1}\varepsilon_{t-1}^\top Q_{t-1}^{*1/2} + bQ_{t-1}$ with $Q_t^* = \text{diag}(Q_t)$, show that $\mathbb{E}[Q_t] = \bar{Q}^*$ exactly

A5: DCC Algebra: Task (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- The four short proofs and the half-life in days; interpretation: what does the half-life say about how fast a DCC hedge ratio drifts, and what else decides how often to rebalance?
- Model: B1

A6: Attainable Correlations: Task (1/2) [Proposed]

Solution discussed in class

Context and data

- ▣ $X = e^{\sigma_1 Z_1}$ and $Y = e^{\sigma_2 Z_2}$ with $Z_1, Z_2 \sim N(0, 1)$: the lognormal margins are fixed, the dependence between Z_1 and Z_2 is free
- ▣ Tools: Hoeffding's identity $\text{Cov}(X, Y) = \iint [F(x, y) - F_1(x)F_2(y)] dx dy$ and the Fréchet–Hoeffding bounds $W \leq C \leq M$

What to do

- ▣ (a) Show that $\rho(X, Y)$ is maximal for $Z_2 = Z_1$ (comonotone) and minimal for $Z_2 = -Z_1$ (countermonotone)
- ▣ (b) Derive $\rho_{\max}$ and $\rho_{\min}$ in closed form from the lognormal moments
- ▣ (c) Evaluate both for $\sigma_1 = 1$, $\sigma_2 = 2$

A6: Attainable Correlations: Task (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- The two formulas and the two numbers; interpretation: what does this imply for reading a Pearson correlation of 0.3 between two price levels?
- Model: A2

B1: DCC for SPY and TLT: Task (1/2) [Solved]

Context and data

- Data: daily adjusted closes of SPY (SPY.US) and TLT (TLT.US) in data/market, prices joined on common days first, then log returns in %; 31 July 2002 – 18 September 2026, $T = 6\,073$
- Model: GARCH(1,1) with constant mean for each series, then DCC(1,1) with correlation targeting (Gaussian quasi-likelihood)

What to do

- (a) Fit GARCH(1,1) to each series and compute the standardised residuals $\varepsilon_{it} = (r_{it} - \hat{\mu}_i) / \hat{\sigma}_{it}$
- (b) Estimate (a, b) by maximising the correlation part of the likelihood; report Hessian standard errors and the likelihood ratio against CCC ($a = b = 0$), with its p -value from a parametric bootstrap of both steps under CCC (199 paths)
- (c) Re-estimate the two steps on 2002–2019 and on 2020–2026

B1: DCC for SPY and TLT: Task (2/2) [Solved]

What to do

- (d) Test for a change in the correlation of ε_t at an *unknown* date with Wied, Krämer & Dehling (2012) (Bartlett bandwidth $\lfloor T^{1/4} \rfloor$, 5% level) and date the break
- (e) On the same 199 residual-bootstrap paths, compute standard errors with step 1 held fixed (step 2 only) and with GARCH re-estimated on each path (both steps); compare both sets with the Hessian ones (Newey & McFadden (1994))

What to report and interpret

- A table of $(a, b, a + b, \bar{Q}_{12}, T)$ for the three samples; the likelihood ratio; the break statistic, its date and the two correlations; the three sets of standard errors
- Interpretation (1): is the persistence $a + b$ stable across subsamples, and what does a moving $\bar{Q}$ mean for a full-sample DCC?
- Interpretation (2): why is the χ^2 reference only indicative for the likelihood ratio, and why is a Fisher z test at January 2022, a date read off the lecture chart, not valid?

B1: DCC for SPY and TLT: Solution (1/3) [Solved]

Solution

- ▣ (a) GARCH(1,1), returns in %: SPY $\omega = 0.0283$, $\alpha = 0.127$, $\beta = 0.848$; TLT $\omega = 0.0057$, $\alpha = 0.049$, $\beta = 0.943$
- ▣ (b) Step 2: $\hat{a} = 0.0522$ (se 0.0059), $\hat{b} = 0.9335$ (se 0.0084), $\hat{a} + \hat{b} = 0.9857$, target $\bar{Q}_{12} = -0.248$
- ▣ Likelihood ratio: $2(\ell_C^{DCC} - \ell_C^{CCC}) = 2(460.3 - 193.5) = 533.4$
- ▣ χ_2^2 is not the null law: under $a = 0$ the parameter b is not identified (Davies problem) and the criterion is two-step; bootstrap under CCC (both steps, $B = 199$): largest null statistic 15.7, $p = 0.005$: constant correlation is rejected
- ▣ (c) 2002–2019: $a = 0.0565$, $b = 0.9199$, $\bar{Q}_{12} = -0.350$ ($T = 4\,386$); 2020–2026: $a = 0.0341$, $b = 0.9527$, $\bar{Q}_{12} = 0.003$ ($T = 1\,687$)
- ▣ Reading: persistence is similar ($a + b \approx 0.98$) but the target moves from about -0.35 to about 0; a full-sample DCC pulls the recent correlation towards a target that no longer applies

B1: DCC for SPY and TLT: Solution (2/3) [Solved]

Solution

- (d) Statistic: $Q_T = \max_j \frac{j}{\sqrt{T\bar{D}}} |\hat{\rho}_j - \hat{\rho}_T|$ with bandwidth $\lfloor 6.073^{1/4} \rfloor = 8$; result $Q_T = 4.25$
- p -value from the Kolmogorov distribution of $\sup |B|$: below 0.001; decision at 5% (1.358): reject constant correlation
- Estimated break 3 August 2020: residual correlation -0.361 before and 0.066 after; testing the two segments again gives a second break on 9 July 2007 ($Q = 3.01$) and $Q = 1.56$ ($p = 0.015$) after the main break
- The fixed-date Fisher test at January 2022 ($z = 14.9$) is not valid: the date was picked after seeing the data, and Fisher assumes i.i.d. Normal pairs
- Economic reading: the stock–bond relation changed in mid-2020, before the 2022 inflation shock, and it changed more than once

B1: DCC for SPY and TLT: Solution (3/3) [Solved]

Solution

- (e) Procedure: decorrelate the residuals with the Cholesky factor of R_t ; resample them; simulate DCC innovations with $(\hat{a}, \hat{b}, \bar{Q})$; (i) re-estimate step 2 only, (ii) rebuild returns with the fitted GARCH and re-estimate both steps
- Standard errors of $(\hat{a}, \hat{b})$: Hessian 0.0059, 0.0084; step 2 only 0.0051, 0.0077; both steps 0.0052, 0.0077
- How to read it: step 1 adds only 2% and 1% because $T \approx 6\,000$ pins down the GARCH parameters; the Hessian understates $se(\hat{b})$ by -8% because the innovations are not Normal
- Conclusion: the inference on (a, b) is robust here, but in short samples or with weak GARCH identification the two-step correction must be made

B2: Gaussian or t Copula for S&P 500 and Euro Stoxx 50?: Task (1/2) [Solved]

Context and data

- Data: the weekly returns and pseudo-observations of A2 (2000–2026, $n = 1\,393$); canonical maximum likelihood (CML) on the ranks
- The Gaussian copula is the t copula with $\nu = \infty$, a point on the boundary of the parameter space

What to do

- (a) Fit the Gaussian and the t copula by CML and compare them by AIC
- (b) Compute the likelihood ratio for $H_0 : \nu = \infty$ and its p -value by parametric bootstrap under H_0 (499 draws)
- (c) Run the Rosenblatt goodness-of-fit test ([Genest, Rémillard & Beaudoin \(2009\)](#)) for both copulas, parametric bootstrap with 499 draws, 5% level

B2: Gaussian or t Copula for S&P 500 and Euro Stoxx 50?: Task (2/2) [Solved]

What to do

- (d) Build 90% parametric-bootstrap intervals for ν and λ (200 draws) and compare $\hat{\lambda}$ with the empirical $\hat{C}(q, q)/q$ at $q = 5\%$
- (e) Compute the standard errors of $\hat{\rho}$ (and $\hat{\nu}$) from the inverse Hessian and from the [Genest, Ghoudi & Rivest \(1995\)](#) sandwich

What to report and interpret

- A table with $\hat{\rho}$, $\hat{\nu}$, ℓ , AIC and the fit p -value; the likelihood ratio with its bootstrap p ; the intervals of (d); the two sets of standard errors
- Interpretation: why not use the χ^2_1 distribution for the likelihood ratio? Does passing the fit test prove that the t copula is right? Why do the ranks add variance?

B2: Gaussian or t Copula for S&P 500 and Euro Stoxx 50?: Solution (1/2) [Solved]

Solution

	$\hat{\rho}$	$\hat{\nu}$	ℓ	AIC	fit p
Gaussian	0.771	—	623.3	-1244.6	0.002
Student t	0.775	3.91	674.2	-1344.4	0.874

- (a) The t copula has the lower AIC by 99.8 points: the extra parameter ν is strongly supported
- (b) Likelihood ratio $2(674.2 - 623.3) = 101.8$; bootstrap $p = 0.002$ from 499 draws under H_0 (the smallest possible is $1/500$); decision at 5%: reject the Gaussian copula
- χ^2_1 is wrong because $\nu = \infty$ lies on the boundary; the textbook fix is the mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$, the bootstrap avoids any such approximation
- (c) Fit test: Gaussian $p = 0.002$, rejected at 5%; t $p = 0.874$, not rejected; not rejecting is absence of evidence against the t copula, not proof

B2: Gaussian or t Copula for S&P 500 and Euro Stoxx 50?: Solution (2/2) [Solved]

Solution

- (d) $\hat{\nu} = 3.91$ with 90% interval $[2.98, 4.77]$; $\hat{\lambda} = 0.467$ with $[0.423, 0.518]$
- Empirical at $q = 5\%$: lower 0.517, upper 0.488; both near $\hat{\lambda}$, with a hint of asymmetry that the symmetric t copula ignores
- (e) Formula: sandwich $A^{-1}\Sigma A^{-1}/n$ with $\Sigma = \text{Var}[\dot{\ell}_{\theta} + W_1(U) + W_2(V)]$, against A^{-1}/n for the inverse Hessian
- Results: Gaussian $\text{se}(\hat{\rho})$ 0.0086 versus 0.0138; t $\text{se}(\hat{\rho})$ 0.0115 versus 0.0131; $\text{se}(\hat{\nu})$ 0.56 versus 0.58
- Interpretation: the ranks are estimated, so the Hessian understates the uncertainty of $\hat{\rho}$; ν is driven by the tails, where the ranking error matters little

B3: Clayton or Gumbel for Joint Losses of BET and Euro Stoxx 50?: Task (1/2) [Proposed]

Solution discussed in class

Context and data

- Data: daily closes of BET and the Euro Stoxx 50 (STOXX50E.INDX) from 5 January 2010, joined on common days (the two markets close almost at the same time); $T = 4\,154$
- Filter each series with GARCH(1,1) and work with **losses** $L = -\varepsilon$ and their pseudo-observations

What to do

- (a) Fit the Clayton, Gumbel, Gaussian and t copulas by CML; report AIC and the implied λ_L, λ_U
- (b) Run the Rosenblatt fit test with parametric bootstrap for each family (200 draws; 100 for the t copula) at the 5% level
- (c) Compute the empirical $\hat{\lambda}_U(q)$ and $\hat{\lambda}_L(q)$ of the losses at $q = 5\%$ and 10%

B3: Clayton or Gumbel for Joint Losses of BET and Euro Stoxx 50?: Task (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- ▣ A table by family (parameter, AIC, λ_L , λ_U , fit p) and the four empirical tail coefficients
- ▣ Interpretation: in which corner of the loss copula are the joint crashes, why can Clayton on losses not capture them, and what would a rotated Clayton represent?
- ▣ Model: B2 (fit and test), A2 (parameters and tails)

B4: Forbes–Rigobon for 2008 and 2020: Task (1/2) [Proposed]

Solution discussed in class

Context and data

- Data: 2-day log returns $\ln P_t - \ln P_{t-2}$ of the S&P 500 (source), Euro Stoxx 50 and BET (targets), on common trading days
- Crises: 15 Sep 2008 – 31 Mar 2009 and 20 Feb – 30 Apr 2020; calm: the year before each crisis; effective n = half the number of overlapping 2-day returns (an approximation)

What to do

- (a) Compute ρ_{calm} , ρ_c , δ and ρ^* ; test $H_0 : \rho_c \leq \rho_{calm}$ and $H_0 : \rho^* \leq \rho_{calm}$ with one-sided Fisher z tests at 5%
- (b) Repeat with crisis windows shortened to 31 Dec 2008 and 31 Mar 2020
- (c) Treat δ as random: stationary bootstrap within each window (mean block 10 days, 1 999 draws); report the 90% interval of ρ^* and the bootstrap p for $H_0 : \rho^* \leq \rho_{calm}$

B4: Forbes–Rigobon for 2008 and 2020: Task (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- ▣ A table by crisis and market with δ , ρ_{calm} , ρ_c , ρ_{raw} , ρ^* , ρ_{adj} , plus the bootstrap results
- ▣ Interpretation: where does the raw test find contagion and does it survive the adjustment? Why is δ so much larger in 2020? Which assumption is least credible for a pandemic?
- ▣ Model: A3

B5: Bitcoin and the S&P 500: DCC with Bootstrap Bands: Task (1/2) [Proposed]

Solution discussed in class

Context and data

- Data: daily closes of Bitcoin (BTC-USD.CC) and the S&P 500 (GSPC.INDX) on common trading days since 18 September 2014, $T = 3018$; weekend Bitcoin moves enter the Monday return

What to do

- (a) Fit GARCH(1,1) and DCC(1,1); compute the average conditional correlation before and after 1 January 2020
- (b) Residual bootstrap: decorrelate $\eta_t = L_t^{-1}\varepsilon_t$ (L_t the Cholesky factor of R_t), resample η_t , simulate a DCC with $(\hat{a}, \hat{b})$ and re-estimate; 200 draws; 90% intervals for a and b
- (c) Recompute ρ_t on the original data with each bootstrap (a, b) and draw the 90% band

B5: Bitcoin and the S&P 500: DCC with Bootstrap Bands: Task (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- ▣ The estimates with intervals, the two averages, the share of days since 2020 with the lower band above zero, and the chart
- ▣ Interpretation: is the post-2020 correlation significantly positive? Why does a simple block bootstrap of ε_t bias $\hat{b}$ downwards? Which uncertainty does the band ignore, and how would you add the first-step GARCH uncertainty?
- ▣ Model: B1

B6: A Dynamic t Copula for S&P 500 and Euro Stoxx 50: Task (1/2) [Proposed]

Solution discussed in class

Context and data

- Data: the weekly pseudo-observations of A2 and B2 (2000–2026, $n = 1\,393$); the static t copula of B2 is the benchmark
- Model: GAS t copula (Creal, Koopman & Lucas (2013), unit scaling): $\rho_t = \tanh f_t$, $f_{t+1} = \omega + A s_t + B f_t$,
 $s_t = \partial \ln c(\hat{u}_t, \hat{v}_t; \rho_t, \nu) / \partial f_t$, $f_1 = \omega / (1 - B)$

What to do

- (a) Derive s_t from the t-copula density and estimate (ω, A, B, ν) by maximum likelihood
- (b) Compute the likelihood ratio against the static t copula, plot ρ_t and report its minimum and maximum with dates

B6: A Dynamic t Copula for S&P 500 and Euro Stoxx 50: Task (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- ▣ The estimates, the likelihood ratio, the range of ρ_t and the chart
- ▣ Interpretation: does the dependence move, and when is it lowest? Why is χ^2 not the reference for this likelihood ratio? B2 did not reject the static t copula: is that a contradiction?
- ▣ Model: B2 (fit), lecture slide on the dynamic t copula

C1: Did Romanian and Euro-Area Equities Become More Integrated after 2020?: Task (1/2) [Proposed]

Solution discussed in class

Context and data

- Data: BET and the Euro Stoxx 50, weekly (Friday) returns on common trading days, 2010–2019 against 2020–2026; daily data for a robustness check
- 2020 brought COVID-19 and the FTSE Russell upgrade of Romania to Secondary Emerging ([FTSE Russell \(2019\)](#)): is a higher correlation integration or only the crises inside the sample?

What to do

- (a) Estimate the change in weekly correlation with a 95% stationary block-bootstrap interval ([Politis & Romano \(1994\)](#), mean block 8 weeks, 2000 draws)
- (b) Two separate robustness checks against the same baseline: (i) the Forbes–Rigobon adjustment for the change in Euro Stoxx 50 volatility on the full post-2020 sample; (ii) the unadjusted correlation without the weeks of 20 Feb – 30 Apr 2020
- (c) Compare Kendall's τ , the BET beta on the Euro Stoxx 50, the lower-tail dependence at $q = 10\%$ and a weekly DCC across the two periods

C1: Did Romanian and Euro-Area Equities Become More Integrated after 2020?: Task (2/2) [Proposed]

Solution discussed in class

What to report and interpret

- A one-slide table of all measures in both periods and a verdict with its caveats; why weekly and not daily returns?

C2: Critique an AI Answer [Proposed]

Solution discussed in class

The prompt sent to an artificial intelligence (AI) assistant

- “e holds the GARCH-standardised residuals of JPM and BAC (two US bank stocks). Calibrate a t copula with $\nu = 4$ from Kendall's τ and give me its tail dependence.”

The assistant's answer

```
import numpy as np
from scipy import stats
tau = stats.kendalltau(e['JPM'], e['BAC'])[0]
rho = 2 * np.sin(np.pi * tau / 6)
u = e.rank() / len(e)
z = stats.t.ppf(u, df=4)
nu = 4
x = np.sqrt((nu + 1) * (1 - rho) / (1 + rho))
lam = 2 * stats.t.cdf(-x, df=nu)

# Kendall's tau
# copula correlation
# pseudo-observations
# input of the t-copula likelihood

# tail dependence
```

- “With $\hat{\tau} = 0.50$: $\rho = 0.518$ and $\lambda_L = \lambda_U = 0.276$.”

C2: Critique an AI Answer, the Task [Proposed]

Solution discussed in class

- Context: the code above runs on the GARCH-standardised residuals of JPM and BAC (daily, 2010–2026), but the answer contains **three** mistakes typical of copula modelling; everything else is correct
 - ▶ Hint: what is u for the largest residual, and what does `stats.t.ppf` return there?
- (a) For each mistake, name the line and explain in one sentence why it is wrong
- (b) Say how you would detect it: a step-by-step check, a formula from A1 or A2, or a small simulation (state which)
- (c) Write the corrected line(s) and recompute ρ and λ for $\hat{\tau} = 0.50$, $\nu = 4$
- (d) Log each mistake in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version
- Report: the three corrected lines and the corrected numbers; interpretation: which of the three errors changes λ the most, and why?

C3: Large Dynamic Covariance Matrices, a Project Seed: Task (1/3) [Proposed]

Solution discussed in class

Context and data

- The case study of Lecture 6 ([Engle, Ledoit & Wolf \(2019\)](#)) shrinks the correlation target $\bar{Q}$ of a DCC model nonlinearly (DCC-NL); in the 1986–2015 backtest with $N = 1000$ stocks the annualised SD of the global minimum-variance (GMV) portfolio falls from 10.59% (sample target, DCC-S) to 8.02%, but with $N = 100$ only from 13.36% to 13.17% ([revised working paper \(April 2017\)](#), Table 7)
- Data: the daily value-weighted returns of the 100 portfolios formed on size and book-to-market and of the 49 industry portfolios ([Kenneth R. French Data Library](#)), 1 January 1990 – latest; a series with a missing value (coded –99.99) in a window is excluded from that window, as the complete-history rule of the paper (Section 6.1)

What to do

- (a) At the start of every “month” of 21 trading days, use the last $T = 1250$ days: fit a GARCH(1,1) model to each series, devolatilise the returns and estimate the target $\bar{Q}$ in two ways, the sample correlation matrix (DCC-S) and its nonlinear shrinkage rescaled to a unit diagonal (DCC-NL, Section 3.5 of the paper)

C3: Large Dynamic Covariance Matrices, a Project Seed: Task (2/3) [Proposed]

Solution discussed in class

What to do

- (b) Estimate (a, b) by composite likelihood over the $N - 1$ contiguous pairs (Section 2.3 of the paper), build the one-day-ahead covariance $\hat{H}_t$ and the GMV weights $\hat{w} = \hat{H}_t^{-1} \mathbf{1} / (\mathbf{1}^\top \hat{H}_t^{-1} \mathbf{1})$ for DCC-S, DCC-NL and CCC-NL (constant correlation with the shrunk target), then hold the number of shares fixed for the 21 days
- (c) Report the annualised mean, SD and information ratio (mean/SD) of the daily out-of-sample returns of $1/N$, DCC-S, DCC-NL and CCC-NL, and test equal SD of DCC-NL and DCC-S with the HAC test of [Ledoit & Wolf \(2011\)](#), as in Section 6.2 of the paper

C3: Large Dynamic Covariance Matrices, a Project Seed: Task (3/3) [Proposed]

Solution discussed in class

What to do

- (d) Change the concentration ratio $c = N/T$: repeat (a)–(c) with the 49 industries ($c = 0.04$) and with the 100 portfolios and $T = 250$ ($c = 0.4$), and compare the mean $\hat{a}$, $\hat{b}$ of DCC-S and DCC-NL, as in Tables 3–4 of the paper

What to report and interpret

- A table with mean, SD and information ratio for each method and each c , the p -value of the SD test and the SD reduction of DCC-NL relative to DCC-S; one chart of this reduction against c
- Interpretation: does the gain of DCC-NL grow with c , as in Tables 2 and 7? If CCC-NL is as good as DCC-NL, what does this say about the value of the correlation dynamics compared with the value of the target?
- Model: B1 (GARCH filtering and DCC estimation), C1 (reading a difference together with its uncertainty); the shrinkage of Lecture 4

Common Pitfalls

- ▣ Computing returns before joining series, or using daily correlations across time zones
- ▣ Testing a break at a date read off a chart; standard errors that ignore estimated ranks or the GARCH step
- ▣ Using χ^2 critical values for tests on the boundary ($a = 0$ in DCC, $\nu = \infty$ in the t copula)
- ▣ Fitting copulas to raw returns instead of GARCH-filtered pseudo-observations
- ▣ Treating a non-rejected copula as the true one, or a rejected one as useless: large samples detect small departures, and a rejection does not say which feature (e.g. time variation) is missing
- ▣ Block-bootstrapping a highly persistent process with short blocks: biased persistence and misleading bands

Take-Home Assignment (Team Project)

- ▣ Choose two markets not covered in class (e.g. WIG20 and BUX, gold and Bitcoin, two bank stocks)
- ▣ Estimate EWMA, rolling and DCC correlations; test DCC against CCC; show the correlation in two crises with the Forbes–Rigobon adjustment
- ▣ Fit five copulas to GARCH-filtered pseudo-observations; report AIC, bootstrap fit tests and tail dependence with intervals
- ▣ Log every AI error you catch in `AI_ERRORS.md` (at least three per project)
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: evidence, interpretation, limitations

Further Study on Quantinar

Advanced courses on quantinar.com

- ☐  Measuring Statistical Risk
- ☐  Multivariate Statistical Analysis

Code and data

- ☐ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_06

References (1/4)

- ▣ Aielli, G.P. (2013). [Dynamic conditional correlation: on properties and estimation](#). *Journal of Business & Economic Statistics*, 31(3), 282–299.
- ▣ Creal, D., Koopman, S.J., Lucas, A. (2013). [Generalized autoregressive score models with applications](#). *Journal of Applied Econometrics*, 28(5), 777–795.
- ▣ Demarta, S., McNeil, A.J. (2005). [The t copula and related copulas](#). *International Statistical Review*, 73(1), 111–129.
- ▣ Engle, R.F. (2002). [Dynamic conditional correlation: a simple class of multivariate generalized autoregressive conditional heteroskedasticity models](#). *Journal of Business & Economic Statistics*, 20(3), 339–350.
- ▣ Engle, R.F., Ledoit, O., Wolf, M. (2017). [Large dynamic covariance matrices](#). Working Paper No. 231, revised version (April 2017), Department of Economics, University of Zurich.
- ▣ Engle, R.F., Ledoit, O., Wolf, M. (2019). [Large dynamic covariance matrices](#). *Journal of Business & Economic Statistics*, 37(2), 363–375.

References (2/4)

- ▣ Forbes, K.J., Rigobon, R. (2002). [No contagion, only interdependence: measuring stock market comovements](#). *Journal of Finance*, 57(5), 2223–2261.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). [Statistics of Financial Markets: An Introduction](#), 5th ed. Springer.
- ▣ French, K.R. [Kenneth R. French Data Library](#): 100 portfolios formed on size and book-to-market (daily); 49 industry portfolios (daily). Tuck School of Business, Dartmouth College.
- ▣ FTSE Russell (2019). [FTSE Russell announces results of country classification review](#) (press release, 26 September 2019).
- ▣ Genest, C., Ghoudi, K., Rivest, L.-P. (1995). [A semiparametric estimation procedure of dependence parameters in multivariate families of distributions](#). *Biometrika*, 82(3), 543–552.
- ▣ Genest, C., Rivest, L.-P. (1993). [Statistical inference procedures for bivariate Archimedean copulas](#). *Journal of the American Statistical Association*, 88(423), 1034–1043.

References (3/4)

- ▣ Genest, C., Rémillard, B., Beaudoin, D. (2009). Goodness-of-fit tests for copulas: a review and a power study. *Insurance: Mathematics and Economics*, 44(2), 199–213.
- ▣ Ledoit, O., Wolf, M. (2011). Robust performance hypothesis testing with the variance. *Wilmott Magazine*, 2011(55), 86–89.
- ▣ Ledoit, O., Wolf, M. (2020). Analytical nonlinear shrinkage of large-dimensional covariance matrices. *Annals of Statistics*, 48(5), 3043–3065.
- ▣ Longin, F., Solnik, B. (2001). Extreme correlation of international equity markets. *Journal of Finance*, 56(2), 649–676.
- ▣ Nelsen, R.B. (2006). *An Introduction to Copulas*, 2nd ed. Springer.
- ▣ Newey, W.K., McFadden, D. (1994). Large sample estimation and hypothesis testing. In R.F. Engle, D.L. McFadden (eds.), *Handbook of Econometrics*, Vol. 4, 2111–2245. Elsevier.

References (4/4)

- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). *Are cryptos becoming alternative assets?*. *European Journal of Finance*, 29(10), 1064–1105.
- ▣ Politis, D.N., Romano, J.P. (1994). *The stationary bootstrap*. *Journal of the American Statistical Association*, 89(428), 1303–1313.
- ▣ Wied, D., Krämer, W., Dehling, H. (2012). *Testing for a change in correlation at an unknown point in time using an extended functional delta method*. *Econometric Theory*, 28(3), 570–589.